

Tung Tran (Tom)

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EDUCATION

- **University of Technology Sydney – Bachelor of Computing Science (Honours)** Sydney, AU
Major: AI & Data Analytics | Sub-major: Statistical Analysis Aug 2025 – 2029 (exp.)
 - **Academic Record:** GPA 6.88/7, WAM 90.25; High Distinction in 7 of 8 subjects, incl. Programming 2 (97), Mathematics 2 (95), Introduction to Data Analytics (90), Discrete Mathematics (88).
 - **Awards:** Dean's List 2026; Academic Excellence International Scholarship (30%); IMC Prosperity 4 Finalist (#9 VN).

SKILLS SUMMARY

- **Machine Learning & Statistics:** Regression, gradient descent, custom losses, time series (HAR, GARCH, ARIMA, LSTM), walk-forward validation, hypothesis testing, PLS-SEM
- **AI & NLP:** Retrieval-augmented generation (FAISS, sentence-transformers, LLMs), semantic search, document Q&A
- **Programming & Tools:** Python, SQL, C++, Java; pandas, NumPy, SciPy, scikit-learn, pytest, Git, Docker, Tableau

RESEARCH EXPERIENCE

- **University of Technology Sydney** Sep 2026 – Present
Research Volunteer – Dark Pattern Analysis Sydney, AU
 - **Literature Review:** Reviewing literature on dark patterns (deceptive UI design) to scope research questions.
- **Research Collaborator** Jan 2025 – Present
Cross-University Research Collaborations (Vietnam, UK & Australia) Hanoi, VN
 - **KDCs & GenAI Adoption (RMIT Vietnam & UK; Q2 paper expected):** Co-designed the survey; ran PLS-SEM (CR, AVE, HTMT, VIF, bootstrapping), t-tests and common-method-bias checks on n=741 responses.
 - **GenAI & Sustainable Innovation (Vietnam & Australia; Q1 paper expected):** Ran PLS-SEM with moderation and MGA on n=989 responses plus NVivo-coded interviews; surfaced a “Too Much of a Good Thing” effect.

INDUSTRY EXPERIENCE

- **Ededge** Nov 2025 – Feb 2026
Data Analyst Intern Sydney, AU
 - **SQL Analytics:** Translated 10 business requirements into SQL queries (GROUP BY, HAVING, UNION) to identify high-value customers, top-performing cities and product/brand revenue contribution across transactional sales data.
 - **Statistical Analysis:** Analysed 2018–2023 daily prices of Microsoft, Apple and Tesla vs. the S&P 500: returns, descriptive statistics and CAPM beta via OLS.
- **Apatek Communication & Technology** Feb 2024 – Aug 2024
Software Engineer Intern Hanoi, VN
 - **LLM & RAG Prototyping:** Prototyped a FAISS + LLM chatbot for automated document Q&A in client support.
 - **Software & Data Systems:** Developed and maintained enterprise applications (10k+ lines of Python/Java), improving runtime efficiency by 20%; designed and deployed a database schema managing 1M+ records.

PROJECTS

- **VN30 Index Futures: Order-Flow Signal & Transaction-Cost Study | [GitHub](#)** Aug 2026 – Present
Personal research project | Python, pandas, PyArrow/Parquet, REST APIs, pytest
 - **Data Pipeline:** Built a 2.6k-line Python package recording live trades and 2-second order-book snapshots via broker REST APIs (no public tick data exists): 320k+ trades and 8 years of bars in Parquet, with 43 pytest tests.
 - **Data Quality:** Designed a reconciliation gate (recorded volume vs. exchange counter; 99.3% cross-source agreement on 32.9k bars) that caught a silent de-duplication bug inflating volume 1.5x; back-adjusted 97 contract rolls.
 - **Modelling & Evaluation:** Engineered a trade-flow imbalance signal, tested it with block-bootstrap CIs and a leave-one-session-out trading rule; showed its +2.35-tick gross edge is erased by ~6 ticks of costs (net -3.7, $t = -3.1$).
- **SPY Volatility Forecasting: HAR Model Trained on QLIKE Loss | [GitHub](#)** Aug 2026 – Sep 2026
UTS Machine Learning project | Python, NumPy, SciPy, pandas, Jupyter/Colab
 - **Model Development:** Extended linear regression to a log-link HAR model trained on QLIKE loss via from-scratch gradient descent; derived the learning-rate stability bound ($\eta < 2/\lambda_{\max}$) and verified it empirically.
 - **Validation Design:** Leakage-free chronological split on 5.4k trading days (2005–2026); tuned the training window on validation years, then retrained daily over a 1,674-day walk-forward test incl. the COVID-19 crash.
 - **Results:** Cut test QLIKE 47% vs. naive and 6% vs. GARCH(1,1) (Diebold–Mariano $p = 0.014$); beat the MSE-trained model in 7/7 years; 99% VaR breach rate 1.79% vs. 2.15% for GARCH.

LEADERSHIP EXPERIENCE

- **Second Chance** Jun 2022 – Present
President Hanoi, VN
 - **Impact:** Raised \$10,000 for STEM and rural-school initiatives; organised STEM sessions for 350+ students.